Emile Dimas

E-mail: emiled16@gmail.com |Phone : +1 (514) 942-0806

Summary

Machine Learning Engineer with 5+ years of experience designing, implementing, and deploying end-to-end production ML systems. Experienced across the full pipeline ranging from data engineering and modeling to system design, I've consistently turned ideas into reliable solutions, including forecasting systems, ML tools and AI agents that drove major customer acquisitions and increased team productivity. Collaborative and multidisciplinary, I am committed to delivering impactful, business-focused results.

Education

Master in Machine Learning - *Université de Montréal & Montreal Institute of Learning Algorithms* **May 2022**

- Graduated with a **4.22/4.3 GPA -** Received a Mitacs scholarship award

Bachelor in Mechanical Engineering - *Polytechnique Montréal* **Dec. 2019**

- Graduated with a **3.98/4 GPA -** Received three scholarship awards

Experience

Machine Learning Engineer/Data Scientist – *Maxa AI* **Jul. 2022 – Present**

- Designed and deployed end-to-end forecasting and ML systems for enterprise clients using advanced feature engineering, segmentation, and regression optimization, deployed as Snowflake-native applications which generated **~$500M in daily operational savings** for a major bank and **$200K/month in cost reductions.**
- Built an internal data science SDK and a Snowflake-native ML platform with fully automated CI/CD pipelines, reducing model development and deployment cycles from **months to days**.
- Developed scalable **DBT** based ETL pipelines on Snowflake with automated CI/CD, integrating data from **10+ systems of record** into analytics-ready, performance-optimized datasets.
- Led adoption of MLOps best practices across the data science team, standardizing tooling, improving reproducibility, and ensuring reliable production deployments.
- Prototyped and shipped internal ML tools and MVPs, including **AI agents** (QA testing, data analysis), **LLM fine-tuning** pipelines on GCP (**SFT**), and automated data discovery tools for analytics engineers.
- Engineered a scalable, fault-tolerant, stateful **MCP server** deployed on GCP, enabling reproducible and guided AI-agent workflows for acquiring data science capabilities.

Machine Learning Intern – *Research Institute of Hydro-Québec* **May 2021 – Apr. 2022**

- Built **distributed Spark pipelines** processing multi-year, high-volume electricity consumption data on a cloud cluster with reproducible workflows.
- Implemented and trained state-of-the-art **deep learning models** for time series forecasting using PyTorch.
- Designed a **reinforcement learning** framework to dynamically select the best forecasting model based on contextual features, improving short-term load forecasting accuracy by 6% over baseline through systematic evaluation and hyperparameter tuning.

Research Assistant – *McGill University - Department of Biomedical Engineering* **Jan. 2020 – Apr. 2021**

- Conducted large-scale modeling on diverse datasets, ranging from genomic cohorts (850K samples) to behavioral data, extracting interpretable patterns in social behavior and brain activity.
- Contributed to peer-reviewed publications through advanced statistical and machine learning analyses.
- Managed the team's data and modeling infrastructure on Google Cloud Platform (GCP), enabling scalable computation, collaboration, and reproducible workflows.

Technical Projects

- Implemented a webscraper SDK orchestrated using **Airflow Helm** charts on a **Kubernetes** cluster hosted on a home lab VMs managed by **Terraform** and set up by **Ansible** playbooks.
- Proposed and implemented **CoopNet**, a deep learning speech enhancement model, integrated into the SpeechBrain framework.

Publications

- "Pattern learning reveals brain asymmetry to be linked to socioeconomic status." *Cerebral Cortex Communications* (in press)
- " The default network of the human brain is associated with perceived social isolation." *Nature communications* (2020)
- "Measuring the dynamic engagement with a system of equations–Theory demonstration and initial analysis." *IEEE (2019)*

Others

- **Skills:** Python, Pytorch, Tensorflow, SciKit-Learn, PyMC3, Pandas, NumPy, Mlflow, Snowpark, Celery, Redis, MongoDB, Postgres, Langchain, FastMCP, Spark, Kubernetes, Helm, Ansible, Terraform, Bash, SQL, Docker, MCP Servers, LLMOps, MLOps, Snowflake, AWS, DBT(data build tool)
- **Interests:** Love to read a Dostoevsky book in the sun after a strenuous hike.
- **Languages:** French, English, Arabic, Russian (learning)